

Predicting NHL Expected Goals (xG): A Machine Learning Approach to Shot Quality

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Presentation Recording Link: [Lewin Marc SAL313 Presentation Recording.mp4](#)

Presentation Slides Link: [Lewin Marc SAL313 Presentation.pptx](#)

1. Executive Summary

The objective of this project is to build an Expected Goals (xG) model to predict the probability of an unblocked NHL shot resulting in a goal. Using play-by-play data from 2018 to 2024, three machine learning models were developed to evaluate shot quality based on physical geometry (angle/distance) and shot type (categorical). The data revealed a severe class imbalance (91% saves/misses vs only 9% goals). To address this, threshold tuning and balanced downsampling were utilized. Ultimately, the Random Forest model performed the best overall, achieving a 64.2% accuracy and the highest F-measure (22.03%). The results prove that shot distance and angle are the biggest drivers of scoring, demonstrating that teams should prioritize shot location over simply trying to tally a high raw shot volume.

2. Problem Description

2.1 Background

Hockey is a low scoring sport where traditional statistics often fail to tell the whole story. A team might outshoot their opponent 40 to 20 but still lose the game. This happens because not all shots are created equal; a 50-foot slap shot from the blue line has a completely different probability of going in compared to a 10-foot tip in from the crease. Therefore, evaluating true offensive and defensive effectiveness requires moving beyond basic shot counts to measure the actual quality of each scoring opportunity.

2.2 Sports Goal and Machine Learning Goal

The sport analytics goal is to objectively quantify shot quality in the NHL, allowing teams to better evaluate player performance and offensive systems.

The machine learning goal is to build a binary classification model that calculates the probability (a continuous value between 0 and 1) that an unblocked shot will result in a goal. While the target variable is binary (*is_goal*: 1 = goal, 0 = no goal), the model's true output is a percentage reflecting the specific danger of each shot.

3. Data Description

3.1 Data

The dataset was sourced using the hockeyR API, consisting of five seasons of NHL play-by-play data (2018-2024, excluding the shortened 2020-21 season). The final dataset contains over 330,000 observations of unblocked shots. The features used to predict the target variable (*is_goal*) include:

- *shot_distance*: Distance from the net in feet.
- *shot_angle*: The angle of the shot relative to the center of the goal line.
- *clean_shot_type*: The categorical type of shot taken (e.g., Wrist Shot, Slap Shot, Deflected, Tip-In). This feature was engineered by grouping messy raw text into clean categories (ex: "wrist" and "Wrist" were originally different shot types)

3.2 Exploratory Analysis

Initial exploration of the data provided clear insights into NHL scoring environments:

- **Density & Location:** The vast majority of shots are concentrated in the high-danger slot and faceoff circles. Shot frequency plummets outside of 40 feet. (See Appendix Figure 1 & 3)
- **Distance Impact:** The median distance for a successful goal is 17 feet, while the median distance for a non-goal is more than double that at 34 feet. (See Appendix Figure 4)
- **Shot Type Efficiency:** Batted pucks, deflections, and tip-ins are scored at significantly higher rates (nearly 15-20%) compared to standard wrist or slap shots (around 5-7%). (See Appendix Figure 2)
- **Angle:** Shooting percentages naturally fade as the angle moves further away from dead center (0 degrees). (See Appendix Figure 5)

3.3 Data Pre-processing

Several steps were taken to prepare the data for modeling:

1. **Coordinate Folding:** All X and Y coordinates were mapped to absolute values to place all shots in a single offensive zone to eliminate negative values and simplify the process.
2. **Filtering:** Blocked shots and empty-net situations were removed to avoid artificially inflating goal probabilities.
3. **Categorical Grouping:** The original, messy shot strings (slap, wrist, snap, etc.) were grouped into 10 clean categories.
4. **Chronological Split:** The data was split chronologically to simulate a real-world scenario. Seasons 2019-2023 were used for the training set, and the 2023-24 season was held out exclusively for the unseen test set.

4. Predictive Modeling (Machine Learning) Solution

4.1 Models

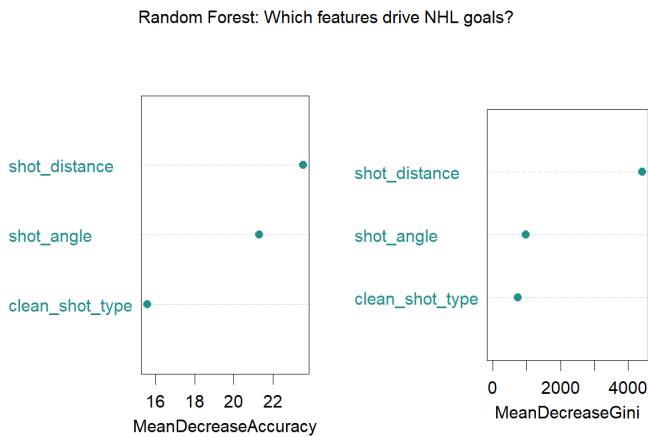
In an NHL game, a goal only happens on about 8.5% of all unblocked shots. Machine learning models usually default to a 50% probability threshold, meaning if an event doesn't have a 50% chance of happening, the model guesses "No." Because hockey is so difficult, almost no shot has a 50% chance of going in. Without adjusting the models, they would achieve 91% accuracy by simply guessing "No Goal" every single time, which would be useless for analysis. To fix this, three different models were trained and tuned:

1. **Baseline Model (Logistic Regression):** The process started with a standard linear equation. The real-world data (91% saves, 9% goals) was retained, but to stop the model from guessing "No Goal" every time, the prediction threshold was manually lowered to roughly 8.3% (see Appendix Figure 7). This allowed the model to successfully flag dangerous shots.
2. **Model 2 (Decision Tree):** A basic decision tree was initially built, but the algorithm failed to identify any goals due to the severe 91/9 data imbalance. Because simply guessing "No Goal" guaranteed 91% accuracy, the model refused to look for deeper patterns and was content with its 91%. To resolve this, the internal weighting was adjusted to force the algorithm to treat predicting a goal and predicting a save as equally important. Finally, cross-validation was utilized to prune the tree down to just 3 highly significant splits (resulting in 4 distinct classification nodes), preventing the model from memorizing random noise. (See Appendix Figure 6 for the Pruned Tree)
3. **Model 3 (Random Forest):** A random forest uses 100 decision trees voting together. To fix the imbalance and speed up computer processing, "balanced downsampling" was used. Instead of giving the algorithm all 330,000 shots, it was given exactly 30,000 goals and a randomly selected 30,000 saves. By training on a perfect 50/50 split, the model naturally learned exactly what a goal looks like.

4.2 Performance Evaluation

All final metrics were calculated using the unseen 2024 test set. ROC curves were used to mathematically find the optimal classification threshold for each model to ensure a fair comparison. (See Appendix Figure 7 for the ROC Comparison)

Model	Accuracy	Precision	Recall	F-Measure
Logistic Regression	61.61%	12.55%	77.84%	21.61%
Decision Tree	59.95%	11.96%	76.88%	20.70%
Random Forest	64.19%	12.93%	74.41%	22.03%



Feature importance analysis from the Random Forest confirmed that shot distance and shot angle are more predictive of a goal than the categorical shot type. The Random Forest proved to be the best overall model, balancing the highest accuracy and F-measure by smoothing out the rigid boundaries of the single Decision Tree. The Logistic Regression model achieved the highest raw Recall, making it highly aggressive at identifying goals, but slightly less accurate overall.

5. Conclusion

5.1 Recommendations

Based on the data and modeling results, NHL teams should prioritize offensive systems that penetrate the inner slot. The models heavily penalize shots taken from beyond 33 feet or at angles wider than 50 degrees. Therefore, players should be discouraged from taking low-danger shots from the blue line just to generate shot volume. Furthermore, the EDA confirms that deflections and tip-ins are highly efficient. Offenses should emphasize establishing net-front presence to create these high probability shot types.

5.2 Limitations

The primary limitation of this project is the reliance on public play-by-play data, which lacks crucial contextual features. The models currently suffer from low precision (~12-13%). This occurs because an open wrist shot from the slot and a heavily contested wrist shot from the slot look identical in this dataset, but the latter is easily saved. The data does not include pre-shot puck movement (like cross ice passes), traffic screening the goaltender, or shooter speed.

Another limitation is the mathematical rigidity of certain algorithms. For example, a Decision Tree draws blocky lines during its process. It might classify a shot from 32.9 feet as highly dangerous, but a shot from 33.1 feet as completely harmless. Because hockey is a fluid game with continuous movement, these rigid boundaries can occasionally misclassify shots that fall just a few inches outside the lines.

5.3 Future Work

Future iterations of this project should incorporate proprietary NHL edge player tracking data to account for goaltender positioning and defensive pressure at the exact moment of the shot. Additionally, game state variables such as score differential and time remaining could be introduced to adjust the model for "score effects," where teams play differently when leading versus trailing.

Appendix: Visualizations

Figure 1: Spatial Shot Density Map

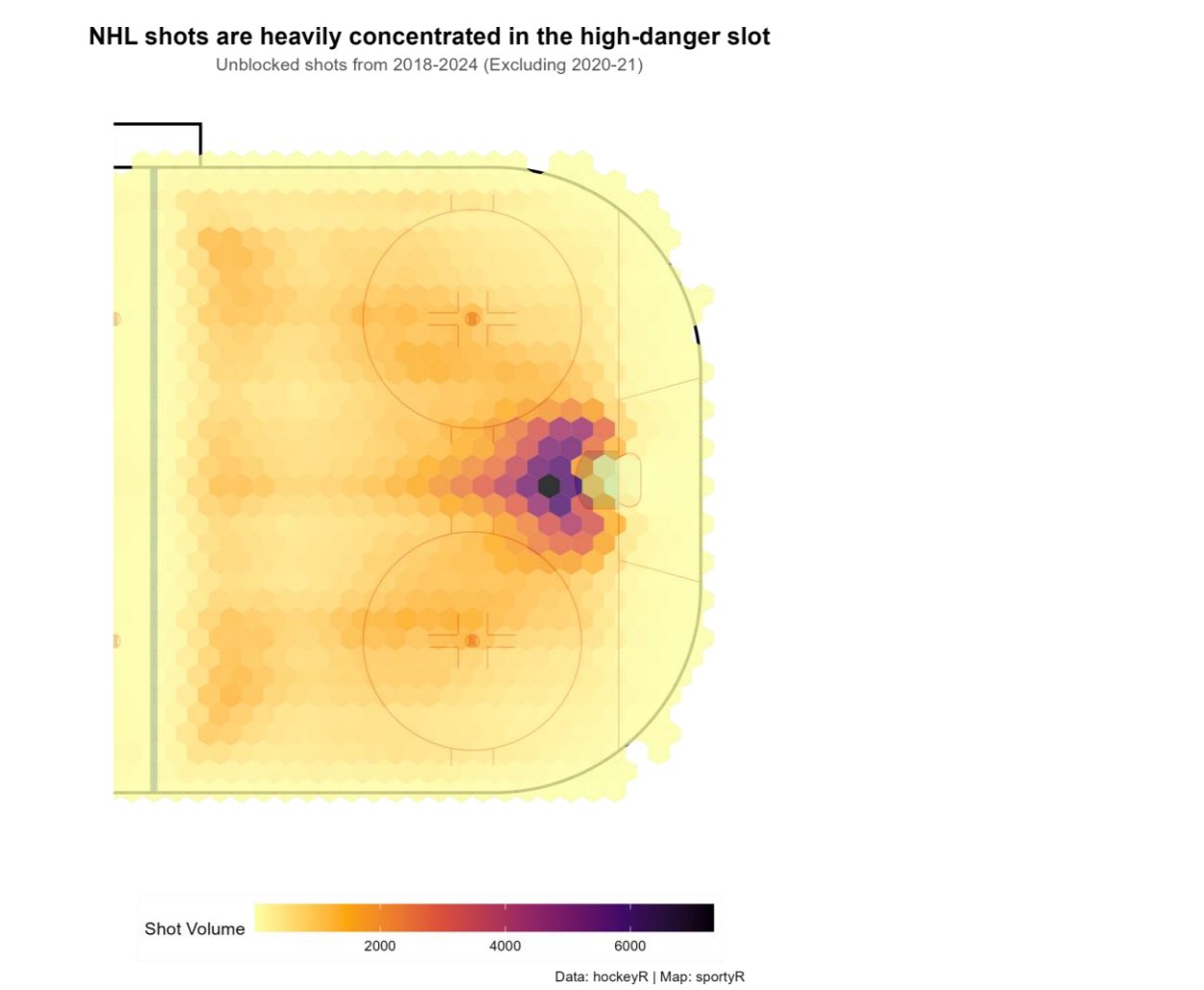


Figure 1: Hex-bin density map illustrating the concentration of unblocked NHL shots (2018-2024). The color gradient highlights the high-danger inner slot area where the vast majority of offensive volume occurs.

Figure 2: Shot Type Conversion Efficiency

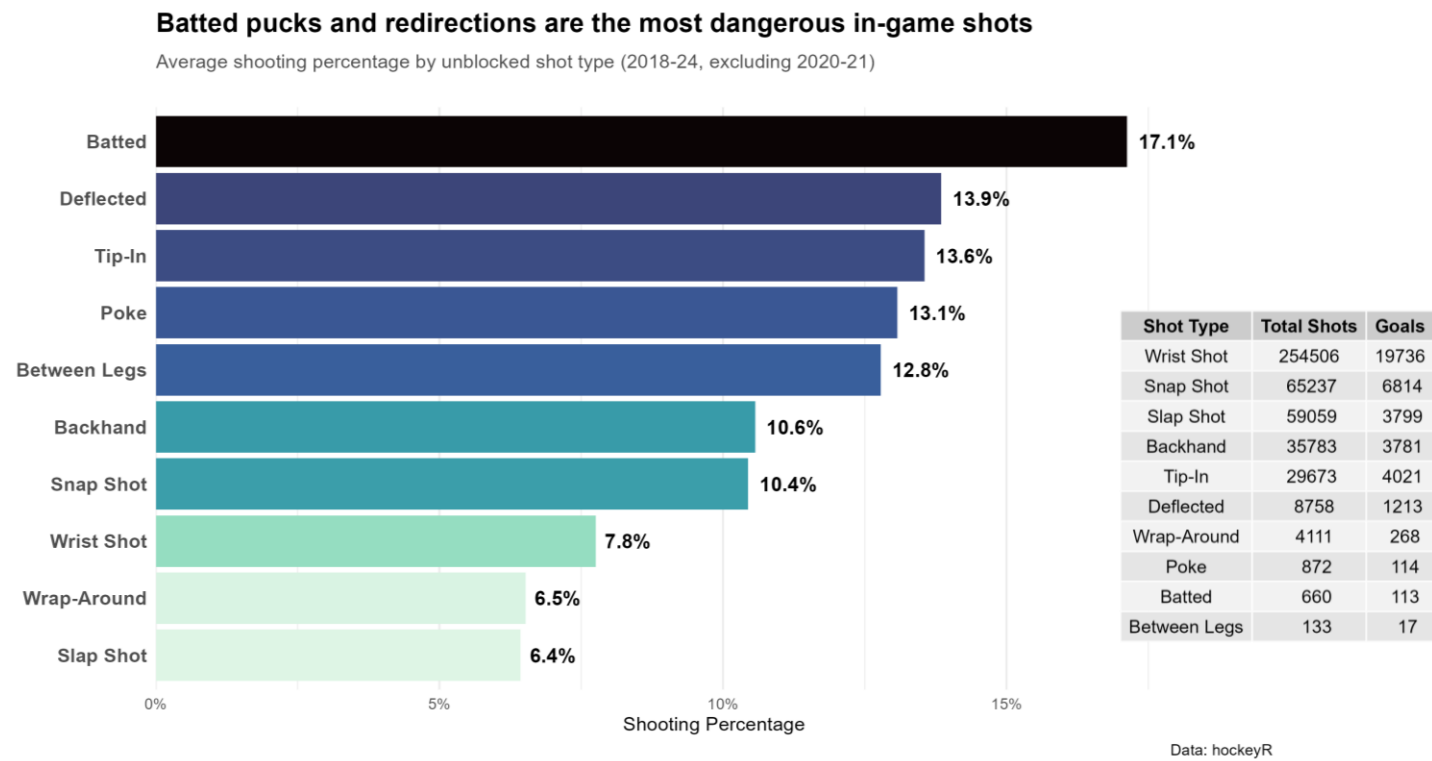


Figure 2: Bar chart displaying the average shooting percentage by shot type. This visualization demonstrates the high conversion rates of complex events like deflections and tip-ins compared to standard wrist or slap shots.

Figure 3: Shot Frequency by Distance

Shot frequency drops dramatically beyond the faceoff circles (40 feet)

Density distribution of unblocked shot distances (2018-24, excluding 2020-21)

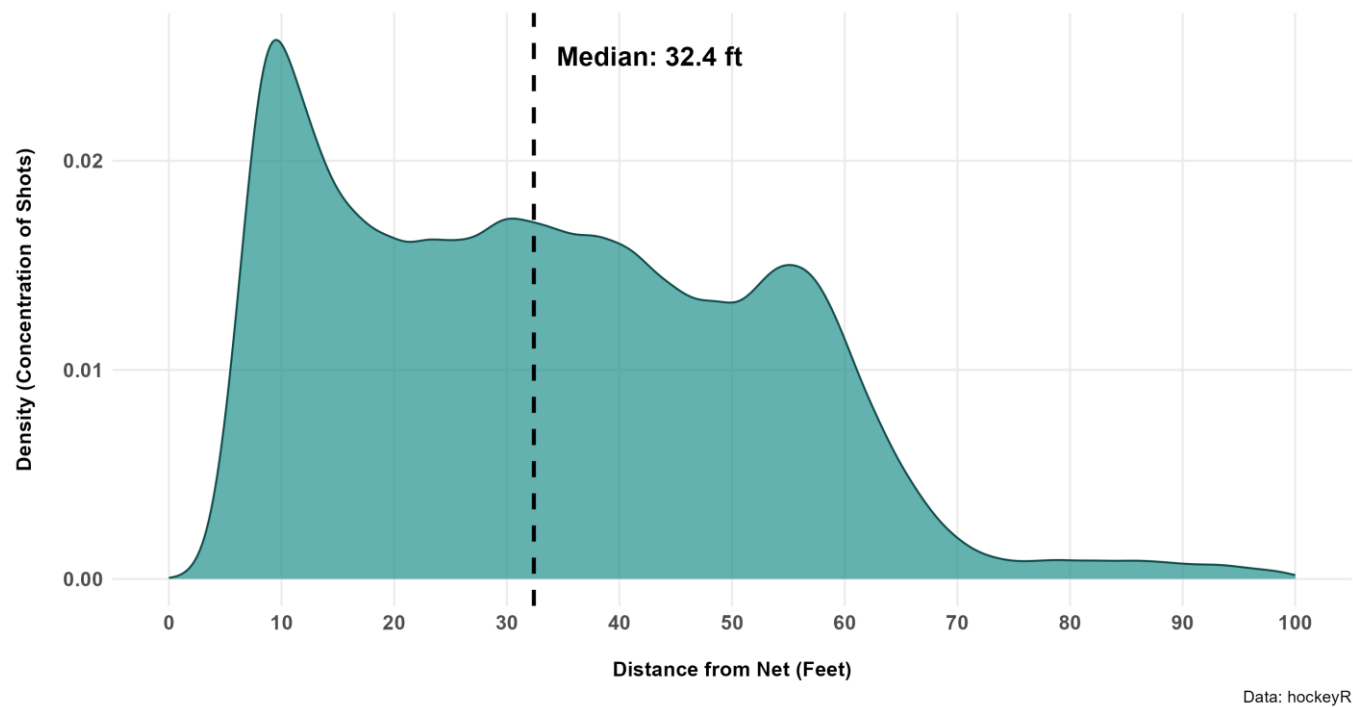


Figure 3: Density distribution curve showing the frequency of unblocked shots relative to their distance from the net. The distribution indicates a severe drop-off in shot frequency beyond the 40-foot mark.

Figure 4: Goal Proximity Comparison

Shot proximity dictates success: Goals occur significantly closer to the net

The median goal is scored from 17.2 feet, while the median non-goal is taken from 33.8 feet.

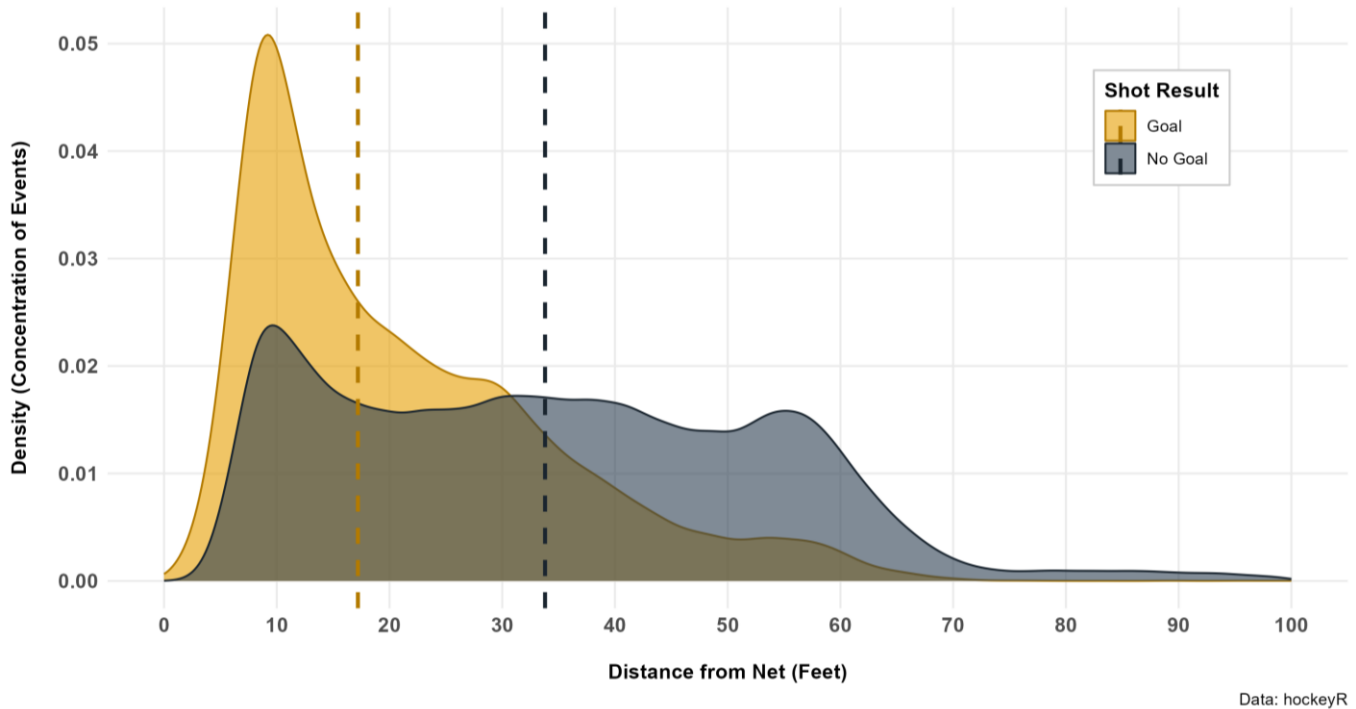


Figure 4: Overlaid density plots comparing the proximity of successful goals versus unsuccessful non-goals. The dashed lines highlight that the median distance for a goal is roughly 16 feet, compared to 34 feet for a save or miss.

Figure 5: Conversion Rate by Shot Angle

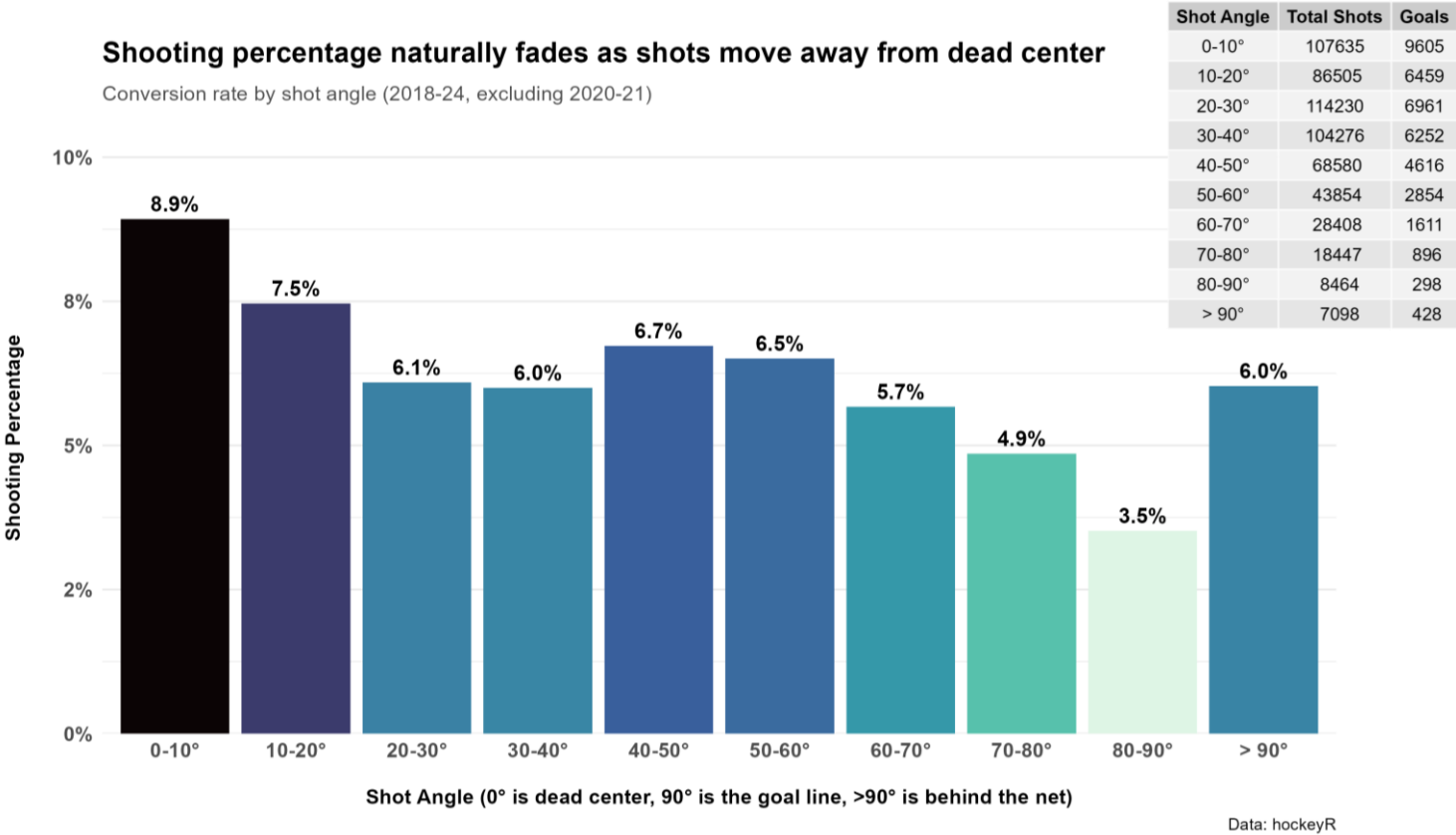


Figure 5: Bar chart illustrating shooting percentage mapped against the shot angle. The data shows a natural and consistent decline in goal conversion as shots move further from dead center (0 degrees).

Figure 6: Pruned Decision Tree Model

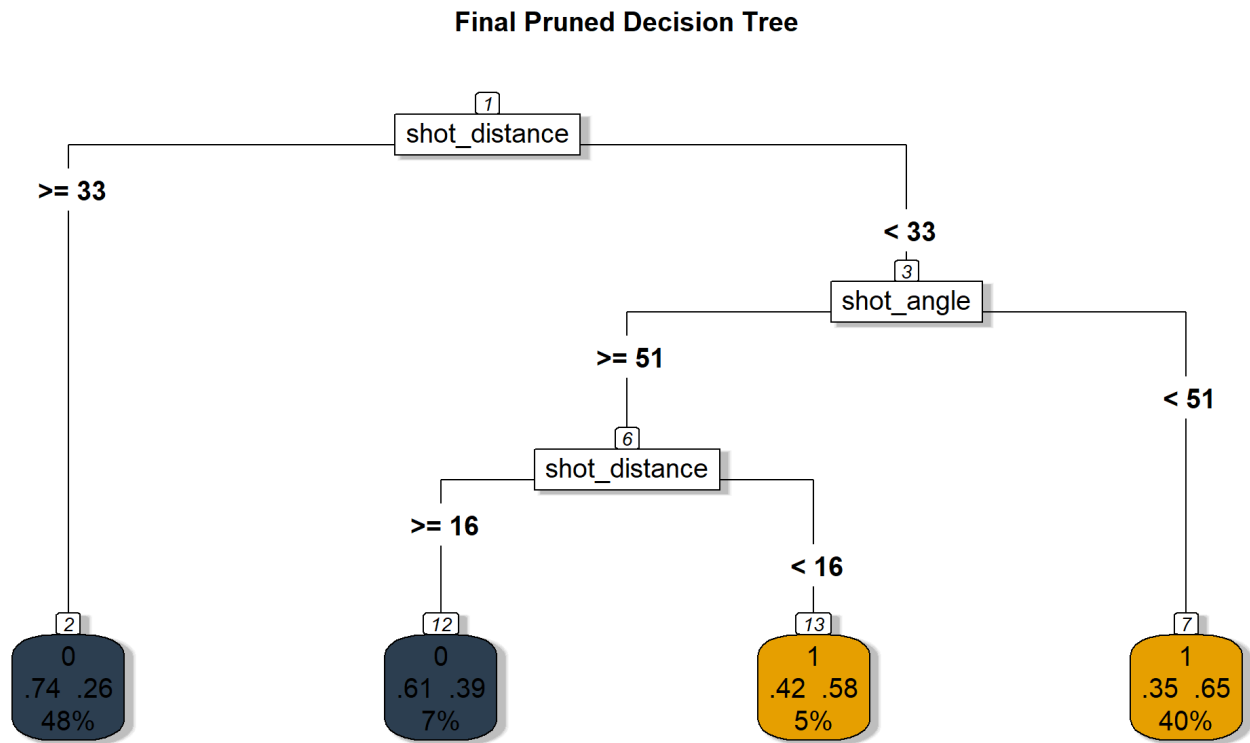


Figure 6: The finalized, pruned Decision Tree model (3 splits). This diagram illustrates the rigid, hierarchical logic the algorithm utilizes to classify a shot based strictly on distance and angle thresholds.

Figure 7: ROC Curve and Threshold Comparison

Why lower the threshold? Finding the analytical sweet spot

A 50% cutoff misses every goal. The 8.3% cutoff captures 78% of them.

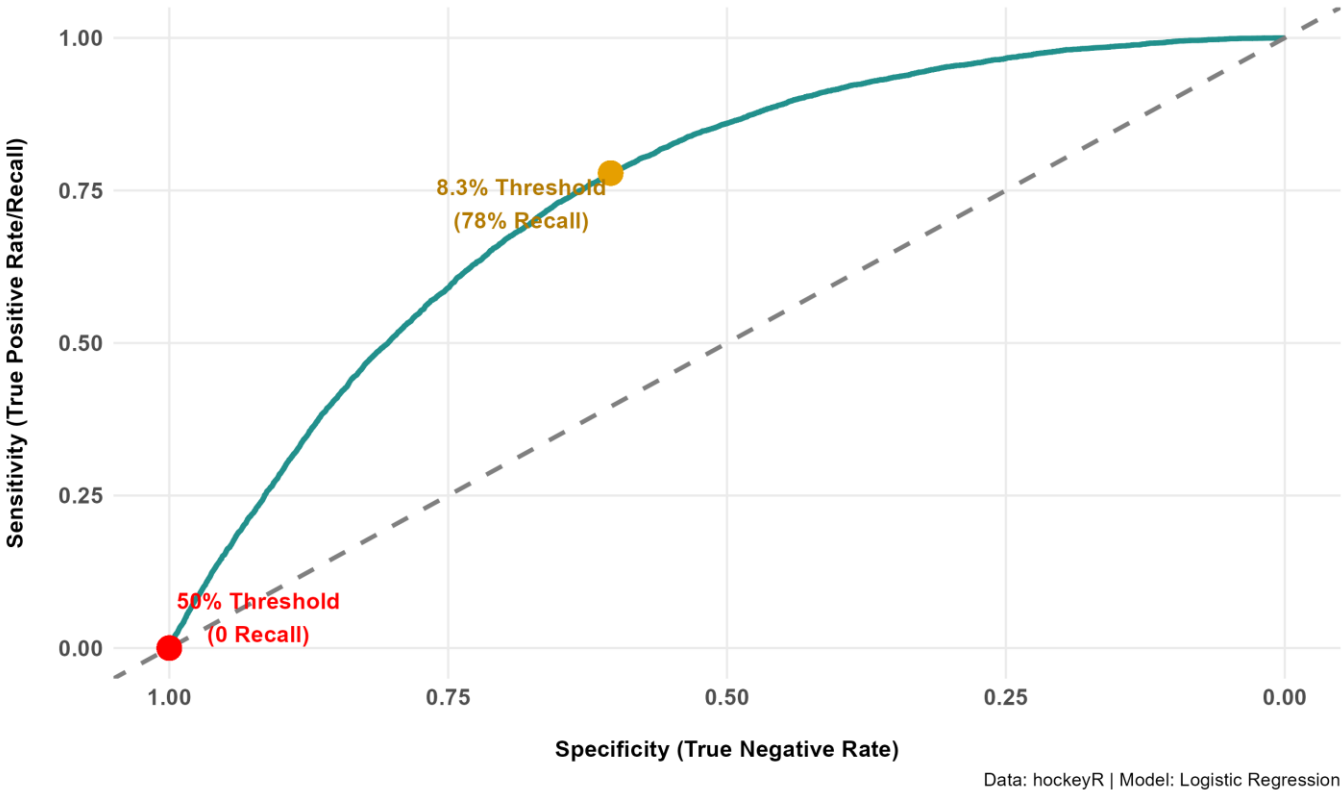


Figure 7: Receiver Operating Characteristic (ROC) curve comparison showing the mathematical trade-off between sensitivity and specificity. This curve justifies the decision to lower the classification threshold to properly capture rare goal events.